

**CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated
for lumbosacral transitional anatomy**

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Purpose: A vertebra at the lumbosacral junction is named by counting. The gold standard counts caudally from C2 on whole-spine imaging¹; however, a lumbar surgical case is planned on lumbar-only imaging²⁻⁴ (T12 to S1) without C2. Abdominopelvic computed tomography (CT) holds that span plus the lowest ribs and pelvis, standing in for the preoperative view. Where a lumbosacral transitional vertebra (LSTV) alters the count, the local anatomy is ambiguous. Four rib-free vertebrae may be an L1 with a lumbar rib or an L5 assimilated to the sacrum, and six may be a sixth lumbar vertebra, a T12 with aplastic ribs, or a lumbarized S1⁵. CTSpinoPelvic1K asks whether local morphology resolves that ambiguity without the count. Two public label sets covered this imaging, CTSpine1K’s vertebrae and CTPelvic1K’s pelvis, on the same colonography patients, never joined; this release joins them on one series and annotates the bones neither had. It provides 802 CT records, per-level ribs and femora, lumbar levels anchored on the lowest rib-bearing vertebra and S1, and classes for L6, T13, a separate S1 and lumbar ribs, so anomalies are recorded as themselves.

Acquisition and Validation Methods: Records pair CTSpine1K⁶ and CTPelvic1K⁷ labels on each patient’s bone-richest acquisition under a VerSe-native scheme. Validation covered geometric invariants (802/802 pass), rib–vertebra incidence across 5,749 ribs (0.035% offset), and spinopelvic measures matching published values.

Data Format and Usage Notes: NIfTI image/label pairs with patient-grouped LSTV-stratified five-fold splits and a loader; archived at <https://doi.org/10.5281/zenodo.22139642>.

Potential Applications: Classifying a vertebra from local features; updating cadaveric morphometry; spinopelvic assessment; opportunistic screening; and, absent a public preoperative lumbar cohort, planning research (377 records prone). *Limitations:* thoracic ground truth is field-of-view limited; postural angles are supine; no held-out test set; ribs are triaged-review pseudolabels; Castellvi grades are two-reader consensus on 33 records.

6 I. INTRODUCTION

7 The spine commonly carries seven cervical, twelve thoracic and five lumbar vertebrae, and
8 a vertebra is identified by where it falls in that sequence. Transitional variants sit at the
9 two ends of the lumbar column, the thoracolumbar transitional vertebra (TLTV) above and
10 the lumbosacral transitional vertebra (LSTV) below, and they disturb identification in two
11 ways at once. They change what the border vertebra and its ribs *look like*: a thoracolumbar
12 border vertebra may carry a full rib, a hypoplastic stump, or none at all, and a lumbosacral
13 one may be partly or completely assimilated to the sacrum (*sacralization*), with an anteriorly
14 wedged body, a reduced disc beneath it and hypoplastic facet joints, or conversely separated
15 from it (*lumbarization*), with a squared body, lumbar-type facets and a full-height disc⁵.
16 They also change *how many* vertebrae each region contains, eleven or thirteen thoracic and
17 four or six lumbar.

18 Either change alone would be tractable. Together they make it difficult to state defini-
19 tively what any vertebra at these borders *is*. Four rib-free vertebrae above the sacrum
20 may mean an L1 bearing a lumbar rib or an L5 assimilated to it; six may mean a true
21 sixth lumbar vertebra, a T12 whose ribs are aplastic, or an S1 lumbarized away from the
22 sacrum. Each set yields the same count (Fig. 2), and the morphology that would separate
23 the readings is not decisive to a human reader in every case. Whether it differs enough for
24 a learned classifier to separate them, or to assign each reading a probability, is the question
25 this release is built to support. Absent that, what remains is the count itself. The gold
26 standard for numeration in transitional anatomy is whole-spine imaging, counting caudally
27 from C2¹, which fixes how many vertebrae and how many rib pairs the column holds, and no
28 spine-limited acquisition can supply either count. LSTV is common, reported at 4–30% in
29 the general population⁵ and in 16.3% of a whole-spine CT series,⁸ and wrong-level surgery
30 is its most serious consequence.

31 Existing public collections cannot address this, and size is not the reason (Table I): spine
32 datasets omit the pelvis, pelvic datasets do not number the vertebrae, TotalSegmentator
33 has no class for a sixth lumbar vertebra or for a rib on a lumbar vertebra, and VerSe, the
34 one collection with an L6 class, stops at the sacrum and carries no ribs. No public collection
35 can therefore record a six-lumbar spine together with both of its borders, and a segmenter
36 trained on them must, faced with an enumeration anomaly, shift the whole column by a level

or absorb a vertebra into its neighbor. The size of that problem has now been measured on CT, where prior labeling methods assigned every vertebra correctly in 77% of subjects and an anomaly-aware extension of SPINEPS⁹ raised that to 99%¹⁰. Its weights label L6 and T13 on CT, but its 1,536-scan CT cohort is in-house and unreleased, so the capability exists as weights, not as data.

The intraoperative side has prior art of its own. LevelCheck registers the intraoperative radiograph to the preoperative CT and projects the CT’s vertebral labels onto it^{11–13}, but those labels are placed by hand and verified by the surgeon¹¹, so the method assumes a correctly labeled CT and does not supply one. Automating that step where the anatomy is anomalous is what a dataset carrying the anomaly classes is for.

CTSpinoPelvic1K places the spine, pelvis, ribs and femora in one coordinate frame and gives a class to each structure an enumeration anomaly produces, a sixth lumbar vertebra and a rib borne by a lumbar vertebra. A scheme without those classes lacks a class vocabulary expressive enough to describe anatomic anomalies.

The project began with a surgeon’s question, whether a vertebra can be named from the anatomy in front of the reader, on the images a lumbar surgical case is actually planned on. By guideline those are lumbar-only imaging studies^{2,3,14}, T12 to S1, the span the ACR spine parameters state for the lumbar spine⁴, and whole-spine coverage is reserved for trauma with an identified injury^{15,16} and for deformity radiographs. An abdominopelvic CT runs from the diaphragm to the pelvic floor, so it holds that span together with the lowest ribs and the whole pelvis, and it lacks C2 exactly as the planning study does. Every record comes from one prospective trial protocol, ACRIN 6664^{17,18}, which scanned 802 patients aged 50 and over, supine and prone, on multidetector CT at 15 centers, on five scanner vendors and nine models, with manufacturer, model and reconstruction kernel recorded per record. To our knowledge it is the largest spine-and-pelvis annotated CT cohort acquired under a single protocol, where the comparators pool collections, are multi-site by design, or are routine clinical scans under many protocols (Table I). With the protocol held fixed, scanner effects on a model become measurable rather than confounded. The price is an abdominal field of view, in which only the lowest thoracic levels are present (Sec. V).

The gap was in plain sight. CTPelvic1K⁷ and CTSpine1K⁶, released months apart by an overlapping author group, each annotated the COLONOG collection under radiologist supervision, one the pelvis and the other the vertebrae, on the same patients, and no one

69 joined them, although the join needs no new imaging and no new radiologist. It is not a
70 file merge, because each annotation had to be traced back to the series it was drawn on
71 (Sec. II.A). Done once, that crosswalk yields a spine-and-pelvis frame at 802 patients, and
72 the bones neither set had, ribs, femora, S1 and hardware, could then be annotated against
73 it rather than from scratch.

74 I.A. Vertebra labeling when the scan cannot settle the count

75 *The limitation, stated plainly.* No scan in this corpus contains C2, so the conventional
76 count cannot be performed, a property of abdominal imaging rather than of the annotation.
77 A thirteenth thoracic vertebra and a lumbar rib are different phenotypes, distinguished as
78 separate subtypes in cadaveric classifications of the thoracolumbar junction^{19,20}: the first is
79 an additional rib-bearing segment, usually above five lumbar vertebrae, so the column holds
80 25 presacral vertebrae; the second is a rudimentary rib on a lumbar-type L1 in a column
81 of the usual 24. Between the two borders used here, the last rib-bearing vertebra and the
82 sacrum, a lumbar rib still leaves four rib-free vertebrae, as does a T13 in the rarer column of
83 24 with four lumbar vertebrae; a T13 above five lumbar vertebrae leaves five, and an aplastic
84 twelfth rib leaves six. A *stump rib*, a hypoplastic twelfth rib and a cardinal indicator of a
85 thoracolumbar transitional vertebra, is still a rib, so it leaves the count unchanged and
86 only its length records the anomaly. Rib anomalies travel with the lumbosacral border;
87 on whole-spine CT, hypoplastic twelfth ribs occur with sacralization and lumbar ribs with
88 lumbarization⁸. Without C2, a count of four therefore cannot separate a lumbar rib from
89 a cranially shifted T13 or from an L5 assimilated to the sacrum, and a count of six cannot
90 separate an aplastic twelfth rib from a sixth lumbar vertebra.

91 *What the release does in addition.* Every vertebra carries the identifier its radiologist-
92 sourced annotation assigned, corrected where the source was wrong; those labels are the
93 ground truth. Every released measurement is also expressed against the two landmarks
94 present in essentially every abdominal scan, the sacrum below and the lowest rib-bearing
95 vertebra above, so that the measurement is positional rather than nominal: a vertebra by
96 how many rib-free vertebrae separate it from the sacrum, a lowest rib by its length as a
97 fraction of the rib above it, a transverse process by its span and its distance from the ala.
98 None of these changes with the name a reader gives the junction, so cases that disagree

TABLE I Public CT collections at the lumbosacral junction. Black, class present; outlined, graded only to exclude it²⁷.

Collection	Scans	Count	L6	Sacrum	S1	Pelvis	Ribs	L. rib	Femora	Grade
CTSpine1K ⁶	1,005									
CTPelvic1K ⁷	1,184									
VerSe ²⁸	374									
RibSeg v2 ²⁹	660									
TotalSegmentator ³⁰	1,204									
CTSpinoPelvic1K	802									

99 about the name remain comparable.

100 *But the vertebrae themselves are known to differ.* Level-specific morphometry is long
101 established: pedicle width and height change systematically from the thoracic to the lumbar
102 spine,²¹ vertebral body, endplate and canal dimensions differ by level,^{22–24} and the transverse
103 process and the iliolumbar ligament mark the last lumbar vertebra independently of any
104 count.^{5,25} If those differences hold at the boundary itself the phenotype is decidable locally,
105 and on this corpus it does. Separating T12 from L1 on shape alone gives an area under the
106 curve of 0.990 and 97.3% accuracy over 1485 vertebrae: a logistic regression on five released
107 per-level measures and six ratios between them, each measure divided by the same patient’s
108 median across their levels, so that size, which separates thoracic from lumbar trivially and
109 says nothing at the junction, is removed; folds are grouped by case and the curve is taken
110 on pooled out-of-fold scores (`test_morphometric_separability.py` in the released code).
111 Schinz et al. reach the same conclusion from the other side: on 1,242 whole-thoracolumbar
112 CTs a shape-based labeling of the junction matched nerve morphology in every case, against
113 92.6–97.2% for counting- and rib-based rules.²⁶

114 II. ACQUISITION AND VALIDATION METHODS

115 II.A. Sources, and why a crosswalk was necessary

116 Both sources draw part of their imaging from The Cancer Imaging Archive (TCIA) CT
117 colonography collection (COLONOG)^{17,18}. CTSpine1K’s 1,005 volumes come from four

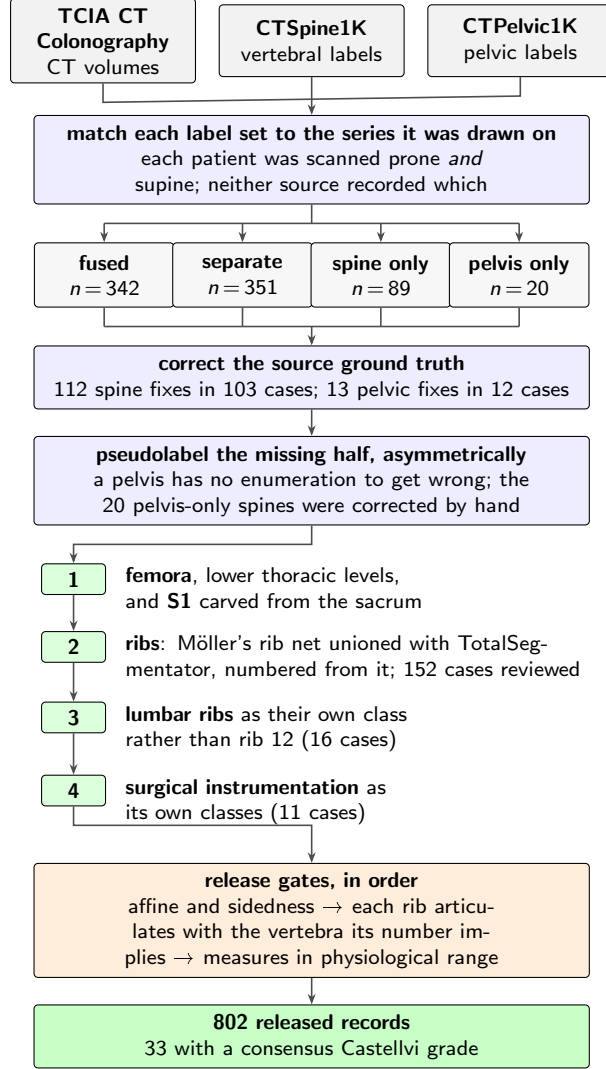


FIG. 1 How the dataset was built. n counts scans, or corrections, at each step.

collections and CTPelvic1K's 1,184 from seven; COLONOG is the only collection common to both and the only one acquired under a single protocol, meaning one scan prescription and patient preparation for every case, so this release is its COLONOG subset, 782 patients with a CTSpine1K vertebral annotation and 20 with a CTPelvic1K pelvic annotation only, 802 in all. Neither source released the mapping from an annotation to the specific CT series it was drawn on.

That mapping matters because every patient in CT COLONOGRAPHY was scanned *twice*, prone and supine, and an annotation carries a patient identifier and nothing finer, so neither collection recorded which acquisition it had labeled. Turning a patient from supine to prone alters lumbar lordosis and the alignment of each segment against the next, so outlines drawn on one series are the wrong *shape* for the other and no rigid realignment recovers

129 them. The crosswalk (Fig. 1) therefore resolves each annotation to a patient and then,
130 within that patient’s volumes, to the series whose bone *agrees* with it, where candidates
131 are thresholded at bone attenuation and scored by overlap with the mask. It mattered for
132 351 patients, nearly half the cohort, whose two collections had labeled different acquisitions
133 of the same person; those records are released but held out of any analysis that assumes a
134 single posture, two labeled postures of one patient being a resource rather than a defect.

135 II.B. Densification, and why it is asymmetric

136 A corpus in which half the records lack a pelvis is not a spinopelvic dataset, so the missing
137 structures were pseudolabeled, asymmetrically (Fig. 1). Pelvis onto spine-only records is the
138 safe direction, since the sacrum and innominate bones carry no enumeration to get wrong,
139 and quality is reported as held-out performance on the *pelvic-only* records, withheld from
140 training for that purpose. The reverse direction carries the whole problem this dataset
141 addresses, because a pseudolabeled spine must commit to a count, and on a transitional
142 vertebra it commits to the commoner reading; those 20 records were corrected by hand.

143 II.C. Ribs

144 No public rib annotation covers this cohort, and the two available automatic sources^{30,31}
145 fail in complementary ways.

146 Möller’s binary rib network³¹ reports high accuracy, and its authors show that stump
147 ribs can be classified from a partial field of view; that result concerns the morphological
148 features computed on a rib once segmented. On this corpus the segmentation itself failed
149 on ribs that were *partially outside* the field of view. A rib crossing the boundary of an
150 abdominal acquisition was liable to be missed altogether rather than segmented to the edge.
151 In a colonography cohort the lower ribs are routinely clipped, and those are the ribs the
152 count depends on.

153 TotalSegmentator³⁰ does segment clipped ribs and supplies the per-rib numbering that a
154 binary network cannot. Its characteristic failure is at the other end of the bone, where the
155 most medial portion of the rib (roughly the last tenth to sixth approaching the costovertebral
156 joint) is frequently absent. A rib truncated at its medial end never reaches the vertebra it

157 articulates with, so the rib–vertebra incidence check (the quality-control gate on the rib class
158 labels, which matches each rib to the vertebra its number implies) has nothing to test.

159 The two were therefore unioned (Fig. 1), TotalSegmentator supplying the numbering and
160 the medial-clipped cases, Möller the detection where its output is more complete.

161 The result is a pseudolabel whose review was triaged rather than exhaustive. A label-
162 based rule flagged every record in which one rib bone carried more than one identifier, or in
163 which a rib failed to reach the vertebra it articulates with, since a single bone carrying two
164 numbers is a numbering error whatever the anatomy. The rule selected 152 records; 148 were
165 reviewed and corrected, and four were not and are identified in the release. Reviewers were
166 medical students working through a student research consortium³², trained against a written
167 labeling protocol, with every correction attributed to its annotator. Records passing the rule
168 were not individually inspected; the residual rib–vertebra offsets reported in Sec. II.F are
169 the independent check on what the rule missed.

170 II.D. Label scheme and counting anchors

171 The scheme is VerSe-native. Vertebrae keep their VerSe identifiers (C1–C7 = 1–7, T1–T12
172 = 8–19, L1–L6 = 20–25, sacrum 26, coccyx 27, T13 = 28), and every non-VerSe structure
173 takes a fixed identifier above that range, S1 carved from the sacrum (29), hips (30–31),
174 femora (32–33), thirteen ribs per side (34–46 left, 47–59 right), lumbar ribs (60–61) and
175 surgical hardware (62–68). The space is contiguous. Rib 13 is the true rib of a T13 and is
176 empty in this release, so a thirteenth thoracic vertebra and a lumbar rib are recorded as the
177 distinct phenotypes they are.

178 Two anchors give every record the same positional frame, and Fig. 2 shows both, the
179 lowest rib-bearing vertebra above, **S1** carved from the sacrum below. S1 supplies the sacral
180 endplate landmark that sacral slope and pelvic incidence are measured from, and with both
181 brackets present L5 and L6 are distinguishable where a spine-limited view alone would not
182 be.

183 *How S1 is cut.* The sacrum’s outer boundary is the source annotation and is never altered;
184 S1 is the cranial part of that same bone, separated by a plane. The plane’s normal is the
185 sacrum’s own cranio-caudal axis, made orthogonal to a left–right axis measured directly
186 rather than assumed, so as to remove the patient’s rotation within the scanner, which is not

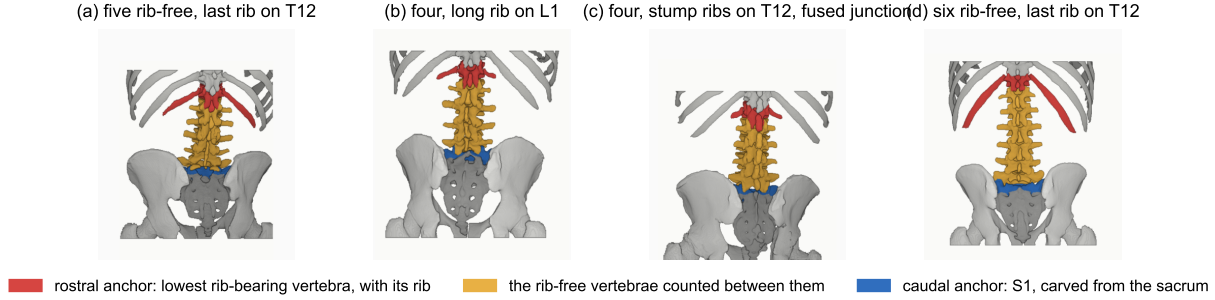


FIG. 2 The two anchors, from the released labels. Red, lowest rib-bearing vertebra and its rib; blue, S1; yellow, the rib-free vertebrae between. (b) and (c) both count four. In (b) a long rib sits on L1; in (c) the twelfth ribs are stumps and the segment below the fourth lumbar body is fused to the sacrum (Castellvi IIIb) and carries the S1 identifier.

negligible: a median of 4.5° , a maximum of 16.1° , and 508 of 801 records above 3° . The cut preserves the sub-division volume of the preceding release, changing the orientation of the S1/S2 boundary and not how much sacrum is called S1; per-record geometry ships with the archive.

VerSe²⁸ carries L6, whose identifier this release keeps, so its L6 *is* VerSe's.

A rib on a lumbar body receives its own class, and this is the one class here with no published counterpart. A lumbar rib and a stump rib are the same object under two counts, so a scheme that numbers every rib 1–12 leaves the annotator either calling it rib 12, asserting the vertebra beneath it is thoracic, the very question at issue, or discarding it. TotalSegmentator carries ribs 1–12 per side and no such class, and VerSe's T13 is a vertebra rather than a rib. This scheme records both: VerSe's T13 (28) with its rib pair (46, 59) for a thoracic-type thirteenth vertebra, and 60/61 for a lumbar-type body bearing a rudimentary rib. No record here carries a T13; the sixteen lumbar-rib cases were read as lumbar-type bodies, and where a border vertebra could not be settled by its readers the label stands as read.

Figure 2 shows why the interval is the primitive and the label is derived. Among the records with four rib-free vertebrae, some reach that count from above, through a rib on L1, and some from below, through a lumbosacral segment fused to the sacrum, in Fig. 2(c) together with stump twelfth ribs; the count alone cannot tell them apart, and the release can because ribs, rib lengths and the S1 boundary are all labeled.

Sixteen records carry a lumbar rib, thirteen bilateral and three unilateral, all on the right,

208 too few to speak to side preference.

209 Eighteen records carry an L6. Seventeen of them carry a consensus Castellvi grade; the
210 eighteenth was not graded. The source transitional label (pelvic where CTPelvic1K flagged
211 one, otherwise the lumbar count) reads lumbarization in fourteen, sacralization in two and
212 semi-sacralization in one, and is unremarkable in the ungraded record. A spine can carry six
213 lumbar-type vertebrae and still be read as a sacralization, depending on where the reader
214 started the count.

215 Both counts, and the manifest fields that report them (`has_l6`, `has_lumbar_rib`,
216 `n_lumbar_labels`, `lumbar_rib_side`), are computed from the released label volumes by
217 counting identifiers 25 and 60–61.

218 II.E. Computational tools

219 *Access.* Every step is performed by open-source code archived with the dataset, together
220 with the reference loader, the label scheme and the quality-control scripts that generated
221 the tables in this manuscript.

222 *Versions and parameters.* Segmentation of femora, the sacral sub-division and the per-
223 level rib numbering used TotalSegmentator³⁰ (`total` task, release 2.x) on a single graphics
224 processing unit. The binary rib network is Möller’s³¹ released weights, applied through the
225 nnU-Net v2 framework³³. The pelvic completion for records whose pelvis was absent used
226 a five-fold nnU-Net v2 ensemble applied out-of-fold, so that no record is completed by a
227 model that has seen it. The five fold checkpoints, with their nnU-Net plans and dataset
228 descriptor, are released alongside the data (the repository URL identifies the authors; it is
229 on the title page and will be restored here for the camera-ready version). Image handling
230 throughout used NiBabel and SciPy; no image is resampled or reoriented when a label is
231 written, so every label shares its image’s grid and affine exactly.

232 *Generative artificial intelligence.* A large language model (Claude, Anthropic) assisted
233 with drafting and editing the manuscript text and with writing the analysis and figure code,
234 all under author review. It was not used to generate, annotate or interpret imaging data;
235 every number reported here is computed by the released code from the released volumes.

236 II.F. Validation

237 Validation has three layers. Geometry is checked first, because a measurement taken on
238 a corpus with a geometric fault still returns a plausible-looking number.

239 *Geometric invariants.* Every case is checked for label geometry agreeing with its CT in
240 shape, affine and voxel spacing; for identifiers within the published scheme; for a non-empty
241 label; and for sided structures falling on the correct sides, with the left–right axis read from
242 the affine rather than assumed.

243 The check compares *each sided pair separately*. Testing the ribs alone passed all 802
244 records, four of which carried a `left_hip` label on the patient’s right side; pooling every
245 sided structure into one test still missed one of those four records, because a large correctly-
246 sided structure masks a smaller transposed one. A gate that passes everything is not evidence
247 of a clean corpus until it has been shown capable of failing.

248 *Rib–vertebra incidence.* Each rib is matched to its nearest vertebra and compared against
249 the vertebra its number implies. Of 11,548 ribs, 5,788 are not evaluable because the expected
250 vertebra lies outside the field of view and 11 have no vertebra within the anchor distance;
251 of the 5,749 evaluable, 5,747 match, two (0.035%) are offset by one level, and no case is
252 misnumbered. The denominator is stated because quoting the two offsets against all 11,548
253 ribs would halve the rate by counting ribs the check never examined. Both offsets are
254 field-of-view truncations, one at +1 level and one at −1.

255 The same check carries an invariant that reads as a null result. Of the 5,749 evaluable
256 ribs, *zero* are assigned to a lumbar vertebra, not because no lumbar ribs exist, since 16
257 records carry one, but because such a rib takes class 60 or 61 and never enters the numbered
258 set the test examines. A numbered rib landing on a lumbar vertebra would be a counting
259 error of exactly the kind this dataset exists to expose.

260 *Agreement with published values.* Derived spinopelvic measures are compared against
261 Vialle *et al.*³⁴ (Table II). Pelvic incidence is the strongest of these checks because it is a
262 morphological property of the pelvis rather than a posture, and so is directly comparable
263 to a standing reference cohort; it agrees to within a degree. Lumbar lordosis sits below the
264 standing reference, the direction recumbency predicts.

TABLE II Derived measures against a standing reference ($n = 260$)³⁴; this cohort is supine.

Measure	This dataset (supine)	Reference (standing)
Pelvic incidence	54.7°	55 ± 10.6
Pelvic tilt	17.3°	13 ± 6
Sacral slope	36.9°	41.0 ± 8.4
Lumbar lordosis	52.7°	60 ± 10
PI – LL mismatch	2.6°	centered near 0

II.G. Castellvi typing

Every record whose vertebral count departs from five carries a Castellvi grade³⁵ in the released manifest, 33 cases typed I–IV with the a/b unilateral–bilateral qualifier. The grade and the count are *different axes*. Grade IIIb occurs here at rib-free counts of four, five and six alike, and seven of the 33 graded cases carry a normal count of five.

Two radiology resident physicians graded every case independently and resolved their six disagreements by consensus; both reads and the consensus are in the manifest. The distinction the readers disagreed on, type II (articulation) against type III (bony fusion)⁵, is the one that matters clinically, and four of the six disagreements were III against II.

II.H. Surgical hardware, and why a threshold is not enough

Metal is trivial to detect and hard to interpret, and the difference matters here because **an iatrogenic fusion is indistinguishable from a congenital one to a distance measurement**. A cage-bridged interspace reads as “no gap” exactly as a fused transitional vertebra does.

II.H.0.a. Detection over-calls by a factor of five. A 2500 HU threshold inside a shell around the labeled skeleton flags 52 of the 802 records. One of the two radiology resident physicians read all 52 and confirmed 11, rejecting 41 (79%) as contrast, calcification and reconstruction artefact. Attenuation does not separate the two groups (both saturate); site and volume do, every confirmed implant at 2,586 mm³ or above and every rejection at 1,768 mm³ or below.

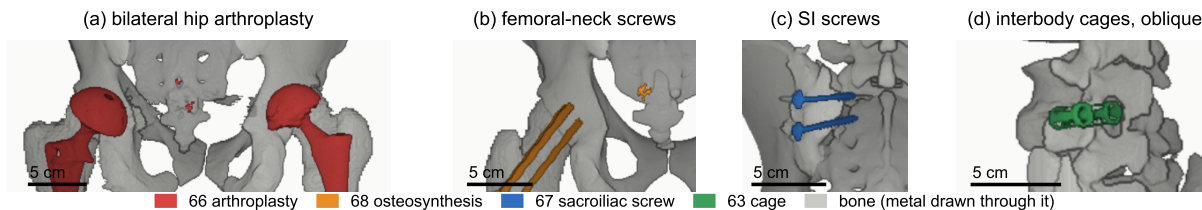


FIG. 3 Instrumentation in the release, one exemplar each, drawn through bone. (a) Hip arthroplasty, 66, $n = 8$. (b) Femoral-neck screws, 68, $n = 1$. (c) Sacroiliac screws, 67, $n = 1$. (d) Interbody cages, 63, $n = 1$. Bar, 5 cm.

285 *II.H.0.b. The subtype block had to be extended.* Identifiers 62–65 name spinal instru-
 286 mentation (generic, cage, screw-and-rod, plate) and cannot describe what this cohort holds,
 287 so **66 arthroplasty**, **67 sacroiliac screw** and **68 osteosynthesis** were added. Osteosyn-
 288 thesis holds parts of one bone together where arthroplasty replaces a joint, and a shape rule
 289 calls a femoral stem a rod. The confirmed cases are eight arthroplasties, one osteosynthesis,
 290 one sacroiliac fixation and one pair of interbody cages (Fig. 3).

291 *II.H.0.c. Metal outranks bone, and that is a subtraction.* In every confirmed case the
 292 implant was already inside a bone label, so naming it reclaims 1,538,852 voxels rather than
 293 adding them. In eight of these cases the femoral head that pelvic incidence and tilt are
 294 measured from is an implant.

295 III. DATA FORMAT AND USAGE NOTES

296 Each record is a gzipped Nifti image/label pair sharing an identical 4×4 affine, canon-
 297 icalized to PIR, requiring no resampling. Masks are Nifti rather than DICOM, which is
 298 what volumetric segmentation tooling expects.

299 Splits are patient-grouped and LSTV-stratified five-fold, frozen and shipped with the
 300 data as `splits_5fold.json`. The five validation folds partition all 802 records, so the
 301 release provides cross-validation and *not* a held-out test set; users reporting a single held-
 302 out number should carve one out and state which records it holds.

303 Stratification is on the transitional subtype rather than a binary LSTV flag. Across all
 304 802 records the source transitional label reads lumbarization in 14, sacralization in 17 and
 305 semi-sacralization in two; separately, 18 records carry a sixth lumbar-type vertebra, nine

TABLE III Data types and metadata, and the records populating each, from the released manifest.

Field	Records Available	
Identifiers and paths, configuration, match type, image/label alignment check	802	100.0%
Patient position, tube potential, section thickness, kernel	802	100.0%
Transitional label and class, annotation origin, instrumentation flags	802	100.0%
Spine series identifier; bone-fraction check	782	97.5%
Scanner manufacturer and model	768	95.8%
Sex	749	93.4%
Age	709	88.4%
Pelvic series identifier	713	88.9%
Castellvi grade (consensus of two readers)	33	4.1%

306 carry only four lumbar labels, and 33 carry a consensus Castellvi grade. A record called
307 sacralization by its four lumbar labels enters the same stratum as one called sacralization
308 by the source label, so the two rare strata hold 15 records each and every validation fold
309 carries three of each. That is the most stratification can guarantee, and it is why a fold-level
310 metric on the rare classes carries an error bar far wider than its own decimal places.

311 The archive of record is Zenodo (<https://doi.org/10.5281/zenodo.22139642>). Code
312 and documentation are on GitHub at the tagged release commit; any model-hub copy is a
313 convenience mirror, not the archive.

314 IV. DESCRIPTIVE ANALYSIS

315 The cohort is a colorectal cancer screening population of 802 records. 738 are recorded
316 female (393) or male (345); 11 carry DICOM’s *other* value and 53 no sex field. Age is
317 present for 709 (median 59, range 50–89). All 802 are counted in totals; the eleven outside
318 the female/male dichotomy are excluded from sex-stratified measures.

319 *Count-free measures* (Fig. 4). Five rib-free vertebrae above the sacrum is typical; four
320 and six are where transitional anatomy sits, and which is present does not by itself determine
321 what to call it. The lowest-rib length ratio is bimodal, and its lower mode, below 0.33 of
322 the rib above, marks a stump twelfth rib in 98 of the 788 records with a measurable pair
323 (12.4%, against 12.6% on whole-spine CT⁸); 16 records carry a lumbar rib (2.0%, against

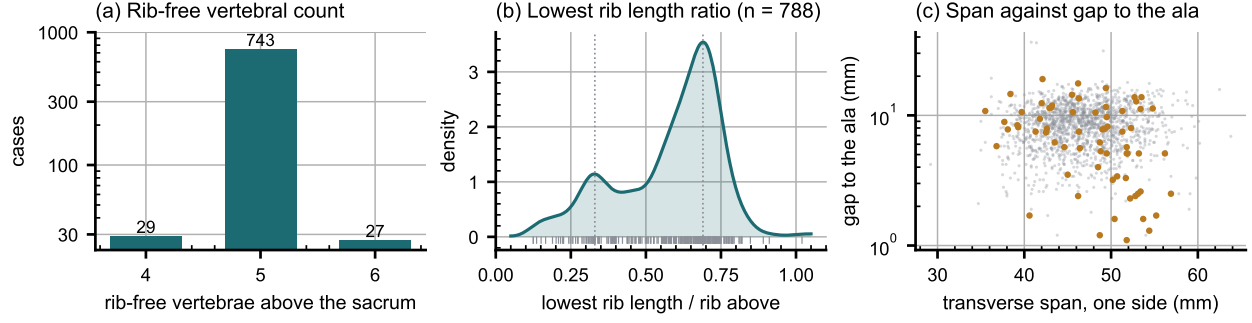


FIG. 4 Count-free measures. (a) Vertebrae between the lowest rib and the sacrum. (b) Lowest rib length over the rib above. (c) Lowest lumbar transverse span against its gap to the ala; transitional labels marked.

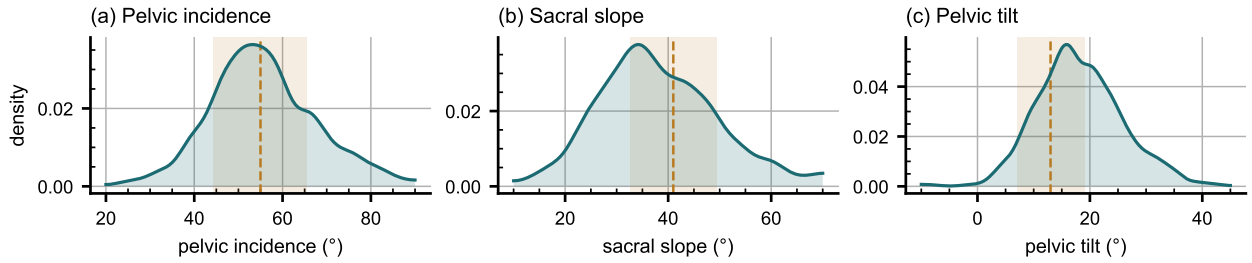


FIG. 5 Spinopelvic measures against standing reference bands (mean $\pm$ SD)³⁴.

324 1.5%). The co-occurrence reported there is testable here only where the junction was read;
 325 stump ribs accompany four of the 14 source-labeled sacralizations and 94 of the other 758
 326 records (odds ratio 2.8, $p = 0.09$, Fisher's exact test), and no lumbarization carries a lumbar
 327 rib.

328 *Level gradients and spinopelvic measures* (Fig. 5). Vertebral dimensions increase caudally;
 329 pelvic incidence does not vary with age while pelvic tilt does, the signature of a pelvis
 330 retroverting as a spine ages.

331 *Opportunistic measures.* Every scan carries a vertebral bone density measurement at no
 332 additional dose.³⁶ L1 trabecular attenuation is reported at the published standard site.

333 V. POTENTIAL APPLICATIONS AND LIMITATIONS

334 The principal use is the *spinopelvic interface*. Spine and pelvis carry radiologist-sourced
 335 annotation on one coordinate frame in 802 records, making sacral morphology measurable
 336 against the lumbar column rather than in isolation, among them the sacral endplate and its

slope, the alar corridors constraining S2-alar-iliac and iliac screw trajectories, and the L5–S1 geometry a lumbosacral fusion must cross. Pelvic incidence is derived from the segmented sacrum and femoral heads, so it is reproducible from the labels themselves.

The transitional layer serves that use rather than competing with it, since no trajectory can be planned across a junction whose segments cannot be named.

A proxy for the preoperative view. No public preoperative lumbar CT cohort exists that we know of, so this release is the closest available stand-in for the view a lumbar case is planned on: it holds the T12-to-S1 span with the lowest ribs and the pelvis, 377 of its 802 records were acquired prone, the position of posterior lumbar surgery, and it carries classes for the anomalies that make level identification fail. It is a screening population at colonography dose, not a surgical series, so it supports method development for level identification and instrumentation geometry rather than validation of a planning decision.

V.A. Three uses the released measurements support

Naming a level from its own geometry. Wrong-level spine surgery runs at roughly one in 3,110 spinal procedures, and among the predominant contributing factors is the transitional anatomy this cohort was assembled around.^{37,38} Per-level body height, canal width, endplate width, transverse-process span and wedge ratio are released for T11–L5 across all 802 records, and because the labels were assigned against the twelfth rib rather than by enumeration, a model trained on them is not learning the convention that fails.

Reference morphometry with its spread. The values a surgeon consults for vertebral body height, canal dimensions and pedicle width come from cadaveric series of twelve to thirty specimens^{22–24} and are redrawn in the textbooks as one curve per level.³⁹ A standard error, where one is reported, describes the mean and not the spread of individuals, so neither tells the reader whether the patient in front of them is unusual. The same measures are given here from 802 records with the distribution attached (Fig. 6), reproducing two properties of the classical figures, body height crossing over at T11–T12 and reversing by L4–L5 and pedicle width widening into the lower lumbar spine, and adding the width of each distribution, the part needed to judge an individual.

Patient-specific models. Finite-element studies increasingly apply patient-specific modeling to alignment planning and implant design.⁴⁰ Such a model needs spine, pelvis and

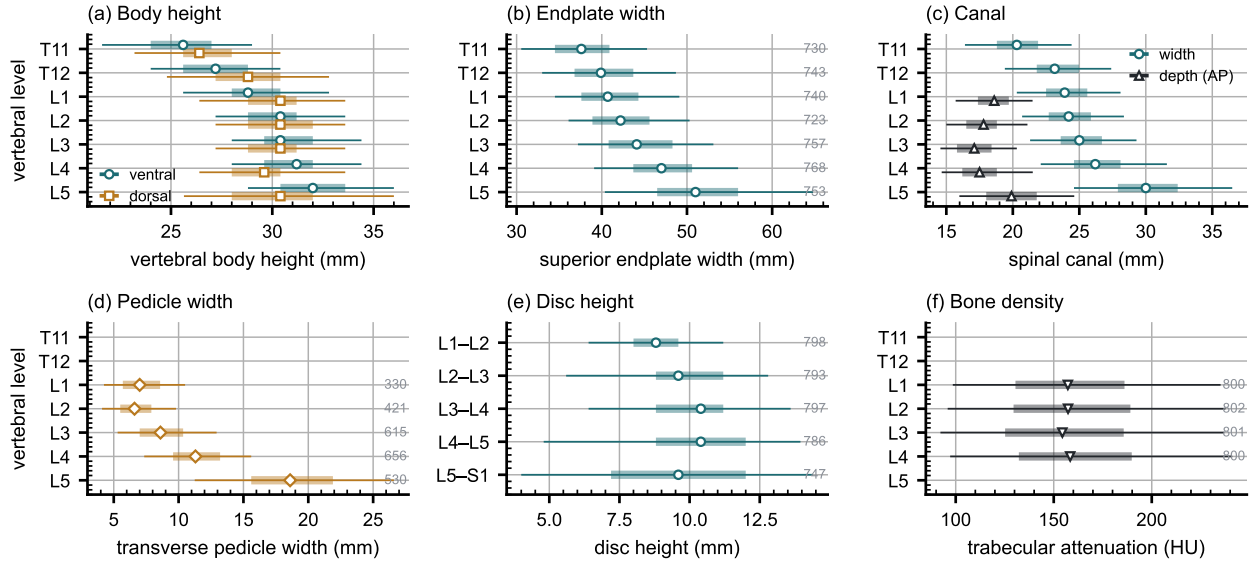


FIG. 6 Morphometry by level, as median, interquartile range and 5th–95th percentile, with n at right. (a) Body height. (b) Endplate width. (c) Canal. (d) Pedicle width. (e) Disc height. (f) Trabecular attenuation.

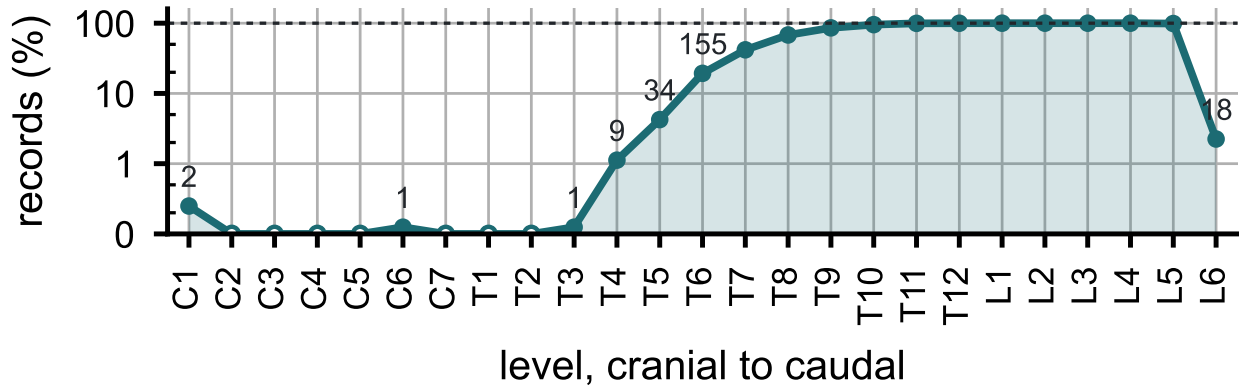


FIG. 7 Records carrying each vertebral level, cranial to caudal. Counts per identifier in Table S1.

367 femoral heads in one frame, which is this release's construction; in 351 patients two postures
 368 give a within-patient change in alignment against which a predicted postural response can
 369 be checked.

370 *Limitations: coverage and cohort.* Thoracic ground truth is field-of-view limited (Fig. 7)
 371 and does not reach T1. Postural angles are supine, though pelvic incidence is not postural
 372 and needs no such caveat. The cohort is one acquisition protocol of a colorectal screening
 373 population aged 50 and over, so its distributions should not be read as a surgical one, and
 374 generalization to pathologic anatomy is untested.

375 *Label strength varies by structure, and the release does not average over that.* Vertebral
376 labels derive from radiologist-supervised source annotations, corrected where those sources
377 were wrong; pelvic labels on records that lacked one are pseudolabeled. The rib layer
378 is a pseudolabel whose review was triaged, so its guarantee is the absence of the failure
379 modes that rule detects. Transitional labels come from two independent sources and are not
380 uniformly adjudicated, which is why the measures reported here are count-free; the Castellvi
381 grades are a consensus of two radiology resident physicians. S1 is carved by an automatic
382 estimate of the S1–S2 boundary; in 100 records that estimate is implausible, making S1 more
383 than half the sacrum’s height or under fifteen percent of it, and those records are flagged
384 in the quality-control table for exclusion from any measurement that depends on the sacral
385 endplate.

386 *Absent structures, and no classifier at the lower border.* Sacrum, hips and femora are
387 present in all 802 records; one record carries no S1 and one no T12, each outside the field
388 of view. Nine records carry no L5 identifier because their source annotation counted four
389 lumbar vertebrae, all nine are Castellvi IIIb, and the fused transitional segment carries the
390 S1 identifier as the caudal anchor; the manifest names them. No shape-based classifier is
391 attempted at the lumbosacral junction, where 15 records in each rare stratum leave three
392 per fold, too few to estimate from; the 0.990 area under the curve reported above is the
393 thoracolumbar boundary, not this one.

394 VI. DISCUSSION

395 *Comparison with related material.* Against the collections in Table I, the contribution
396 is the crosswalk placing spine and pelvis on the same series, together with classes for the
397 anatomy an enumeration anomaly produces. VerSe^{27,28} comes closest in intent but has no
398 sacral mask for the structure a Castellvi grade describes; this release segments it, as L6 or
399 as a carved S1. Extending VerSe with a sacrum would not have supplied the pelvis, femora
400 or prone acquisitions, and what this release adds over TotalSegmentator³⁰ is the classes an
401 anomaly needs, not more anatomy per scan.

402 *Future directions.* Imaging beyond this protocol, and the C2 anchor it cannot supply,
403 are the natural next additions; applying the released weights to VerSe would add the S1
404 anchor and the anomaly classes to a bone-kernel collection. A Castellvi read of the whole

cohort would test the co-occurrence of rib and lumbosacral anomalies at full power and would supply the labels a lumbosacral classifier needs. The S1 carve should be replaced by one that follows the anatomy, and whether the fused segment in the nine four-lumbar records is a lumbar-type body is a question for pelvic morphometrics.

VII. CONCLUSION

CTSpinoPelvic1K places spine, pelvis, per-level ribs and femora on one coordinate frame in 802 records, and gives the anatomy that makes lumbar numbering ambiguous a class of its own rather than recording it as something it is not. Because both counting anchors are explicit, a level can be identified from a field of view that cannot support the conventional count downward from C2, which is every record here.

DATA AVAILABILITY

The release described here is v10, permanently archived at <https://doi.org/10.5281/zenodo.22642578>; the concept DOI <https://doi.org/10.5281/zenodo.22139642> resolves to the current version. The archive carries the loader, the label scheme, the quality-control tables and the analysis code under an open-source licence. The repository URL identifies the authors; it is on the title page and will be restored here for the camera-ready version.

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CONFLICT OF INTEREST

The authors have no relevant conflicts of interest to disclose.

ETHICS

This work uses publicly available, de-identified imaging and annotations derived from it. The authors’ institutional review board determined that it is not human participant research under 45 CFR 46 and requires no oversight.

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543 FIGURE CAPTIONS

544 **Figure 1.** How the dataset was built. n counts scans, or corrections, at each step.

545 **Figure 2.** The two anchors, from the released labels. Red, lowest rib-bearing vertebra and
546 its rib; blue, S1; yellow, the rib-free vertebrae between. (b) and (c) both count four. In (b)
547 a long rib sits on L1; in (c) the twelfth ribs are stumps and the segment below the fourth
548 lumbar body is fused to the sacrum (Castellvi IIIb) and carries the S1 identifier.

549 **Figure 3.** Instrumentation in the release, one exemplar each, drawn through bone. (a)
550 Hip arthroplasty, 66, $n = 8$. (b) Femoral-neck screws, 68, $n = 1$. (c) Sacroiliac screws, 67,
551 $n = 1$. (d) Interbody cages, 63, $n = 1$. Bar, 5 cm.

552 **Figure 4.** Count-free measures. (a) Vertebrae between the lowest rib and the sacrum. (b)
553 Lowest rib length over the rib above. (c) Lowest lumbar transverse span against its gap to
554 the ala; transitional labels marked.

555 **Figure 5.** Spinopelvic measures against standing reference bands (mean $\pm$ SD)³⁴.

556 **Figure 6.** Morphometry by level, as median, interquartile range and 5th–95th percentile,
557 with n at right. (a) Body height. (b) Endplate width. (c) Canal. (d) Pedicle width. (e)
558 Disc height. (f) Trabecular attenuation.

559 **Figure 7.** Records carrying each vertebral level, cranial to caudal. Counts per identifier in
560 Table S1.